A bounded research programme, with its corrections retained

1. Discovery

Search-selected rule; the large initial demonstration used a localised random patch.

2. Confirmation

Frozen seeds reproduced irregular, nonmonotone large-domain growth.

3. Composition

Dense regions are mixed excitations; sparse regions are near-vacuum.

4. Controls

Pure rotation and the active Q+M sibling did not reproduce the same regime.

5. Homogeneous starts

Separate whole-world random ensembles established density dependence.

6. First passage

A geometrical endpoint was followed through longer observation windows.

7. Prepared shapes

Large shapes persisted; relative-growth counts are not absolute formation tests.

8. Intervention audit

Shuffling destroyed measured structure while crossing the legacy detector.

9. Current conclusion

The causal history claim is withdrawn. The observed reversible domain phenomenon remains.

Explanatory timeline of existing studies • no new scientific measurement